Benchmarking Deep Learning Approaches for Splicing Forgery Detection in Document Image Preprocessing Pipelines

Rishikesh Gopal, P Sriya Reddy, Abhi Trivedi, Deepika Gupta
Indian Institute of Information Technology Vadodara–International Campus Diu, India
{202311072,202311060,202311001,deepika_gupta}@diu.iiitvadodara.ac.in

ABSTRACT

A major challenge in the field of document image forensics is the prevalence of diverse digital image manipulation techniques, among which image splicing remains one of the most common and deceptive forms. Accurately identifying such forgeries is essential for applications related to security, legal evidence, and journalistic integrity. Because of the shortcomings in generalization displayed by earlier handcrafted feature based techniques across a variety of datasets, the shift to deep learning based methodologies is recommended. The CASIA 2.0 tampering detection dataset is utilised in this study to carry out a comparative analysis of three leading convolutional neural network architectures, ResNet50, InceptionV3, and DenseNet201. For a 99% detection accuracy with a well-balanced precision and recall, DenseNet201 performs consistently better than ResNet50 and InceptionV3. These results clearly underscore the superior capability of DenseNet201 in the realms of gradient flow and feature extraction, and establishing it is the best option for detecting image splicing effectively. This comparative analysis serves as a critical consideration when selecting deep learning models for forensic applications where both efficiency and reliability are primary considerations.

CCS CONCEPTS

• **Computing methodologies** → **Computer vision**; **Neural networks**; **Image manipulation**; • **Security and privacy** → **Privacy protections**.

KEYWORDS

Image splicing detection, forgery detection, deep learning, convolutional neural networks, transfer learning, ResNet50, InceptionV3, DenseNet201, CASIA 2.0

ACM Reference Format:
Rishikesh Gopal, P Sriya Reddy, Abhi Trivedi, Deepika Gupta. 2025. Benchmarking Deep Learning Approaches for Splicing Forgery Detection in Document Image Preprocessing Pipelines. In *Proceedings of the Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP '25), Dec 17, 2025, Mandi, India.* ACM, New York, NY, USA, 8 pages. https://doi.org/XXXXXXX.XXXXXXX

ICVGIP '25, Dec 17, 2025, Mandi, India

ACM ISBN 978-1-4503-XXXX-X/18/06
https://doi.org/XXXXXXX.XXXXXXX

1 INTRODUCTION

The authenticity verification of digital images has become an increasingly important issue since they are widely utilized in many fields of media, communication, and security. Meanwhile, advanced editing software can easily be accessed, which dramatically increases the number of digital image manipulations and brings serious threats to online safety, credibility, and evidence integrity of all the manipulation methods, such as copy-move, removal, and splicing, the latter is considered the most challenging one since it usually provides a visually plausible appearance and can be integrated without being easily detected.

The rapid proliferation of digital media has greatly impacted the ways in which everyone accesses and shares information. When digital documents are used heavily in every government and official processes. This trend, however, is concurrently occurring with the widespread usage of advanced image manipulation techniques. In order to confirm the legitimacy and authenticity of digital content of the documents, document image forensics, which looks for indications of image manipulation, has grown in importance in recent years.

Among various manipulation techniques, image splicing is arguably the most common and challenging kind of digital forgery. Splicing refers to combining and cutting portions of images from various sources to create a single piece as shown in Fig. 1. In use cases such as faking the ids in banking sector, law enforcement for fabricating evidence, and security for verifying imposter identities, manipulations that are usually invisible to the human eye can have a great impact. Hence, in order to guarantee the integrity of digital document content, reliable detection techniques are needed.

Historically, splicing detection technologies have been based on metadata analysis or statistical analysis, or manually crafted features of irregular color, edge artifacts, and noise patterns [1, 2]. Although such methods tend to be relatively successful, they are always tested in the context of modern forgeries that might be assisted by advanced editing software and post-processing techniques to mask traces of manipulation. Besides, it is hard for traditional methods to generalize to large datasets and real-world applications. As, conventional methods for the detection of forgery prove inadequate in identifying contemporary editing methods and various image compression formats. Through the automatic acquisition of hierarchical features that identify subtle discrepancies in photographs that have undergone modification, deep learning specifically, Convolutional Neural Networks (CNNs) has demonstrated itself to be a potent solution.

In the current study, three advanced CNN architectures, including ResNet50, InceptionV3, and DenseNet201, are compared for splicing forgery detection. Using a standardized dataset, we hope to evaluate these models' performance using a number of performance

Figure 1: Demonstration of Image Splicing - two separate images (laptop and glasses) are combined into a single composite image.

metrics, including accuracy, precision, recall, F1-score, and AUC. The structural advantages of each of these architectures make them advantageous. The current research clearly outlines various merits and demerits related to each approach and also provides positive suggestions regarding the choice of an appropriate model for the digital forensic domain.

1.1 Motivation for Using Deep Learning

Image classification and visual recognition have been transformed by deep learning, especially in the area of convolutional neural networks (CNNs), which allows models to learn rich, hierarchical feature representations from raw pixel data on their own [3]. Unlike traditional handcrafted methods that rely on manually engineered features like color inconsistencies, edge artifacts, or noise patterns, CNNs can detect subtle, high level tampering cues in spliced images that are often undetectable to human observers or traditional forensic tools.

The availability of pretrained deep architectures, which have been trained on large datasets like ImageNet, such as ResNet50, InceptionV3, and DenseNet201, provides a solid foundation for transfer learning in forensic applications. These models are excellent candidates due to their remarkable generalization and resilience across a variety of visual domains. By fine tuning only the top layers on splicing specific datasets, we employ domain adapted feature extractors, which perform noticeably better than classical approaches in both detection accuracy and resistance to post processing attacks (e.g., JPEG recompression, blurring, or resizing) [4]. To find the architecture that provides the best balance of performance, efficiency, and deployability in actual digital document verification pipelines, a comparative forensic study is required.

1.2 Objectives of the Research

This study's primary objective is to perform a comparative evaluation of three cutting-edge Convolutional Neural Network models ResNet50, InceptionV3, and DenseNet201 on the CASIA 2.0 image splicing benchmark dataset [5]. In particular, the study seeks to:

- Using key metrics like accuracy, precision, recall, F1-score, and AUC, measure and compare detection performance to ensure a fair evaluation in the context of class imbalance, which is prevalent in forensic datasets [6].
- Examine the model's resistance to various splicing artifacts, which are frequently used in real-world situations to hide tampering. Sharp edges, blended transitions, color inconsistencies, and JPEG compression are examples of these artifacts.
- Examine computational efficiency in terms of memory usage, inference speed, and training time. These metrics are crucial for deployment in forensic environments with limited resources.
- Examine the effectiveness of architectural features (such as multi-scale processing, residual learning, and density of connectivity) in representing characteristics unique to splicing, including boundary irregularities and spatial patterns of noise.
- Recommend measures for choosing models based on application context, independent of whether the aim is maximum accuracy (for instance, legal evidence), real-time processing (for instance, media verification), or resource-limited execution (for example, devices near the edge).

By giving them clear, actionable insight, this structured comparison aims to assist forensic practitioners in bridging the gap between deep learning performance and real-world deployment constraints. The research also lays the groundwork for future hybrid models and ensemble strategies that might integrate the complementary advantages of these architectures to attain even greater detection reliability in adversarial forensic settings.

The rest of the paper is organized as follows. Literature review of various techniques has been provided in Section 2. The proposed methodology is discussed in Section 3 followed by the experimental results and analysis of the results in Section 4. The conclusion and future work is provided at last in Section 5.

2 LITERATURE REVIEW

Digital image forgery detection research has witnessed the transition from manually developed features to today's advanced methods that employ computer-based approaches. This development conveys the need for more reliable, broadly usable forensic solutions and also highlights the development of investigative techniques becoming more complicated over time. Approaches to image forgery detection using computer-assisted identification and the traditional methods of hand developed features fall into two primary types: CNN (Convolutional Neural Network)-based deep-learning approaches and the more traditional methods that involve human-curated features.

2.1 Handcrafted Feature-Based Methods

Handcrafted features, which searched for structural and statistical differences in forged images, were a major component of early forgery detection techniques. These techniques developed forensic markers that could detect indications of tampering using domain expertise. Among the widely used techniques were:

- **Features of Pixel and Noise Level:** Techniques examined discrepancies in JPEG compression distortions, color filter array (CFA) patterns, or sensor noise. For instance, spliced regions are frequently indicated by local noise variance mismatch or demosaicing artifacts [2].
- **Frequency Domain Analyses**: Methods used transforms like Discrete Wavelet Transform (DWT) or Discrete Cosine Transform (DCT) to reveal aberrant frequency distributions brought on by recompression or splicing [7].
- **Boundary and Edge Analysis**: Gradient based or contour based descriptors can be used to identify spliced regions, which usually introduce artificial edges or boundary irregularities.
- **Statistical Descriptors:** As forensic markers, higher order statistics, texture descriptors (such as Local Binary Patterns (LBP)), and lighting irregularities were employed [1].

Although these methods yielded beneficial results in controlled settings, they had limited generalization, required extensive parameter adjustment, and frequently failed when exposed to complex post-processing operations (such as noise addition, resizing, or blurring) intended to conceal manipulation traces. Their usefulness in operational forensic pipelines was limited by their notable performance degradation across a variety of datasets and real-world situations.

2.2 Deep Learning-based Approaches

Convolutional Neural Networks(CNNs) have revolutionized image forensics by learning rich, hierarchical features directly from raw pixels, dispensing with the need to design hand crafted descriptors that previously dominated the work. CNNs are excellent at picking up on minute variations in texture, noise, and edges that conventional techniques frequently overlook when it comes to splicing detection.

In patch-based CNNs the input image is divided into patches that overlap or do not overlap, and each patch is independently categorized as either tampered with or authentic. This method struggles with global context and boundary blending, despite being effective on localized artifacts.

Transfer learning from pretrained models like ResNet50, InceptionV3, and DenseNet201, which were initially trained on ImageNet, are refined on tampering data because forensic datasets like CASIA 2.0 are not too large. While adjusting to forensic cues, only the last layers are modified, maintaining general visual characteristics.

To detect both fine-grained and large-scale manipulations, some frameworks fuse outputs from multiple kernel sizes or combine spatial domain CNNs with frequency domain inputs (e.g., DCT coefficients).

Forensic reporting and evidence presentation depend on pixel-level tampering heatmaps, which are produced by attention mechanisms, saliency maps, or fully convolutional networks in addition to binary classification [8].

With transfer learning becoming the de facto standard in contemporary splicing detection pipelines, these deep learning techniques have continuously outperformed classical feature engineering [9, 10].

2.3 Comparative Insights

Digital document image forensics has been transformed by the shift from hand designed features to deep learning. Although hand designed methods laid the groundwork with basic techniques, they are no longer regarded as adequate for contemporary, high quality manipulations. Deep Learning techniques, particularly pretrained architectures such as ResNet50, InceptionV3, and DenseNet201, have been iteratively pretrained using large-scale datasets and current state of the art and their current research to quantify their performance trade-offs on benchmark splicing datasets, offering useful insights for forensic applications [11].

3 PROPOSED METHODOLOGY

Traditional approaches of forgery detection rely on manually created features and are not very generalizable. The ability of Deep learning models to learn hierarchical feature representations has revolutionized image analysis. For small forensic datasets, transfer learning that enhances models pretrained in ImageNet is necessary. This study serves as a preliminary stage toward the broader objective of identifying forgeries in document images, focusing specifically on splicing detection as a foundational preprocessing step. In this work, we have used three deep learning models, namely, ResNet50, InceptionV3, and DenseNet201, to detect forgery in the images.

3.1 Transfer Learning

Convolutional Neural Networks (CNNs) are perfect for identifying subtle splicing artifacts because they can learn hierarchical features from raw pixels. In general, in CNN pipeline of splicing detection, early layers identify edges/noise mismatch, mid layers record texture inconsistency, and final layers categorize authentic vs tampered. Here, transfer learning is used to modify ResNet50, InceptionV3, and DenseNet201 for the purpose of forensic splicing detection. So early and mid layers are frozen and final layers are retrained for this dataset.

Here, ImageNet pretrained weights are loaded into every model. Only the top classifier is adjusted on CASIA 2.0, with the base layers remaining frozen. The guided fine tuning of the models is done using the binary cross entropy loss as given below.

$$\mathcal{L} = -\frac{1}{N} \sum \left[y \log \hat{y} + (1 - y) \log(1 - \hat{y}) \right]$$

Table 1 provides the architectural features of different models used in the study. The model architectures are shown in Fig. 2.

Table 1: Summary of Transfer Learning Architectures Used

Model	Key Architectural Features	Output and Classifier Details
ResNet50 (Residual Learning)	• Input: $224 \times 224 \times 3$ • Conv1: 7×7, stride $2 \rightarrow 112 \times 112 \times 64$ • 16 Residual Blocks: $y = F(x) + x$	• Global Average Pooling → 2048-dim • Fully connected layer → 2-class Softmax
InceptionV3 (Multi-Scale Processing)	• Input: $299 \times 299 \times 3$ • 11 Inception Modules: parallel 1×1, 3×3, 5×5 • Grid Reduction layers halve spatial size	• Global Average Pooling → 2048-dim • Fully connected Softmax for 2-class output
DenseNet201 (Dense Connectivity)	• Input: $224 \times 224 \times 3$ • 4 Dense Blocks: $x_l = H_l([x_0, ..., x_{l-1}])$ • Growth Rate $k = 32$, Total Layers = 201	• Output feature vector: 1920-dim • Fully connected classifier for 2 classes

3.2 ResNet50: Residual Learning with Deep Supervision

The problem of vanishing gradients in very deep architectures is resolved by the ResNet50, 50 layer deep convolutional neural network using residual connections. It learns a residual function $F(x) = H(x) - x$ in place of a direct mapping $H(x)$, so that:

$$y = F(x, \{W_i\}) + x$$

This identity shortcut allows for the steady training of hundreds of layers by enabling direct gradient flow [12]. In image splicing detection, ResNet50 excels at capturing fine-grained hierarchical features, early layers detect edge discontinuities and noise mismatch, while deeper layers detect semantic inconsistencies across spliced regions.

After max pooling and a 7×7 convolution (stride 2), the architecture moves on to 16 bottleneck residual blocks (1×1, 3×3, 1×1 convolutions). After being reduced to a 2048 dimensional vector by a global average pooling layer, the feature map is fed into a binary classifier. After being pre-trained on ImageNet, only the final layers are refined on CASIA 2.0.

After performing max pooling and a convolution (stride 2), the model uses 16 bottleneck residual blocks (with each consisted of 1 × 1 and 3 × 3 convolutions). The bottleneck layer produces a feature map which is then reduced to a 2048 dimensional vector by a global average pooling layer before being classified by a binary classifier. The model had been pre-trained on ImageNet, but will only refine the last layers of the model on CASIA 2.0.

3.3 InceptionV3: Multi-Scale Feature Extraction

The 48 layer deep convolutional neural network InceptionV3 uses Inception modules with parallel convolutions of different filter sizes (1×1, 3×3, 5×5) to capture multi-scale contextual information. It reduces computational costs by using filter factorization [13]:

$$\text{Output} = \text{Concat}\left(\text{Conv}_{1\times1}, \text{Conv}_{3\times3}, \text{Conv}_{5\times5}, \text{Pool}\right)$$

An auxiliary classifier, label smoothing, and batch normalization make up the network. For splicing forgery detection, InceptionV3 looks for local tampering patterns in receptive fields.

Once the input images have been resized to 299×299, 11 Inception modules process them and classify them using global average pooling (2048-dim).

3.4 DenseNet201: Dense Connectivity and Feature Reuse

The 201-layer convolutional network DenseNet201 uses dense connectivity, in which all inputs from the layers before it are received by each layer:

$$x_l = H_l([x_0, x_1, \ldots, x_{l-1}])$$

This feature reuse promotes maximum information flow, improves gradient propagation, and reduces parameters [14]. DenseNet201 efficiently captures rich, multi-level features in splicing detection.

The architecture is composed of 4 dense blocks with growth rate $k = 32$, which are separated by transition layers. The final global average pooling reduces the output to 1920 dimensions.

4 EXPERIMENTAL RESULTS AND ANALYSIS

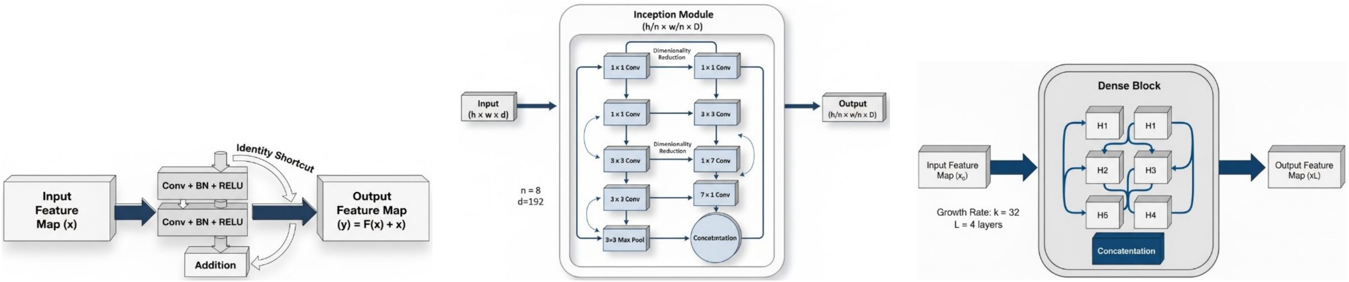
This section presents the experimental result carried out in this study with their detailed analysis.

4.1 Dataset

We used the 7,000 images (2,000 tampered, 5,000 authentic) in the CASIA 2.0 Image Tampering Detection Dataset [15]. For each model, the same dataset was used to ensure a fair comparison. The details of the dataset used are provided in the Table 2.

4.2 Experimental Setup

Each experiment was conducted on Kaggle Notebooks equipped with Intel Xeon CPU (4 vCPUs), NVIDIA Tesla T4 GPUs (16 GB



(a) Resnet50 **(b) InceptionV3** **(c) DenseNet201**

Figure 2: Overview of the deep learning backbones evaluated for splicing forgery detection in document images: (a) ResNet50, (b) InceptionV3, and (c) DenseNet201.

Table 2: CASIA 2.0 Dataset Statistics

Statistic	**Value**
Total Images	7,000
Image Classes	2 (Authentic, Tampered)
Authentic Images	5,000
Tampered Images	2,000
Types of Forgeries	Copy-move, Splicing, Retouching
Image Resolution	512×512 pixels
Format	JPEG
Annotations	Labeled; ground truth masks provided

VRAM), and 25 GB RAM. The components of the software stack are:

- Python 3.9, TensorFlow 2.12.0, Keras 2.12.0
- OpenCV 4.8, NumPy 1.24, Matplotlib 3.7
- scikit-learn 1.3 for metrics and cross-validation

The CASIA 2.0 dataset was split into 80% training and 20% testing sets using 5 fold cross validation to ensure robustness. For reproducibility, a fixed random seed (42) was used to train each model. The input images were normalized using ImageNet statistics and resized to model specific dimensions (ResNet50/DenseNet201: 224×224; InceptionV3: 299×299).

4.3 Hyperparameters

Binary cross-entropy loss is used. Rotation, flipping, and color change are a few examples of data augmentation; the batch size ranges from 16 to 64; the epochs range from 20 to 50; and the learning rate is between 0.0001 and -0.001. The first layers are frozen and the top layers are adjusted.

4.4 Performance Metrics

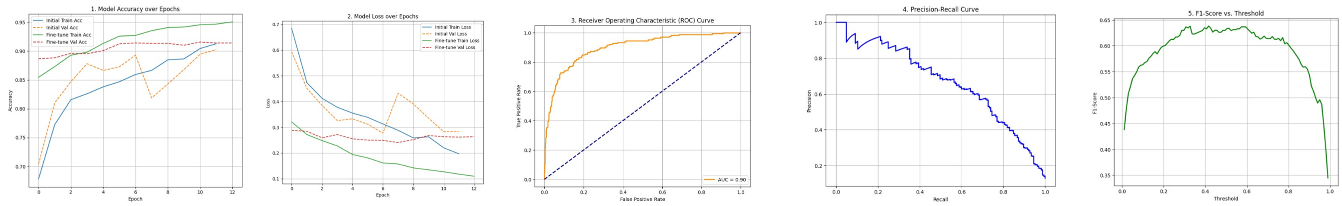
The following common forensic evaluation metrics were used to thoroughly assess the models:

- **Accuracy:** $\frac{TP+TN}{TP+TN+FP+FN}$ — measures the overall correctness of the model.
- **Precision:** $\frac{TP}{TP+FP}$ — represents the proportion of detected forgeries that are actually forged.
- **Recall (Sensitivity):** $\frac{TP}{TP+FN}$ — indicates the model's ability to correctly identify all tampered images.
- **F1-Score:** $2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$ — the harmonic mean of Precision and Recall, suitable for imbalanced datasets.
- **ROC Curve and AUC:** Plots the True Positive Rate (TPR) against the False Positive Rate (FPR) to evaluate classification robustness across varying thresholds.
- **Training and Validation Curves:** Convergence behavior is assessed using accuracy and loss plots across epochs.

4.5 Results

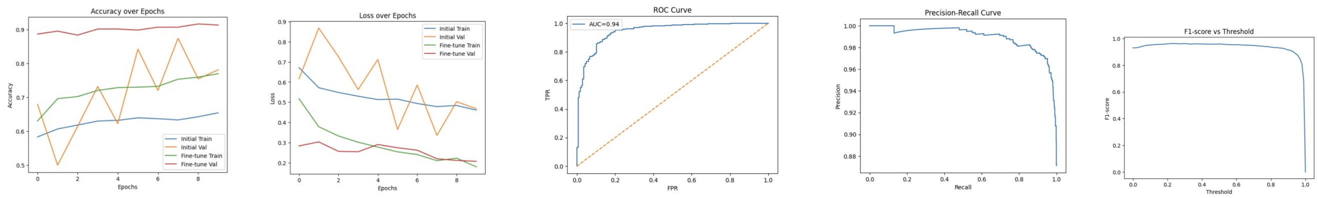
Table 3 provides the quantitative results of the different models considered for the study. The performance is compared against four performance matrices, Accuracy, precision, Recall, F1-Score, and AUC-ROC for a comprehensive assessment. With the highest accuracy of 99.42% and F1-score of 98.24%, DenseNet201 performs better than ResNet50 and InceptionV3. ResNet50 provides strong generalization with 99.40% accuracy, while InceptionV3 performs slightly worse but still maintains a high precision of 97.80%. These results show that DenseNet201 is the best choice for forensic image splicing detection due to its excellent feature reuse and gradient flow. Among all, DenseNet201 shows the best and most reliable results for all the metrics, which supports its choice as an optimal model that can be integrated within the preprocessing pipeline for document image forgery detection.

Figures 3, 4,and 5 illustrate various graphical representations of the experimental results ResNet50, InceptionV3, and DenseNet201 respectively, providing a clearer visualization of model performance across different evaluation metrics. The confusion matrices of all the models are shown in Fig. 6.



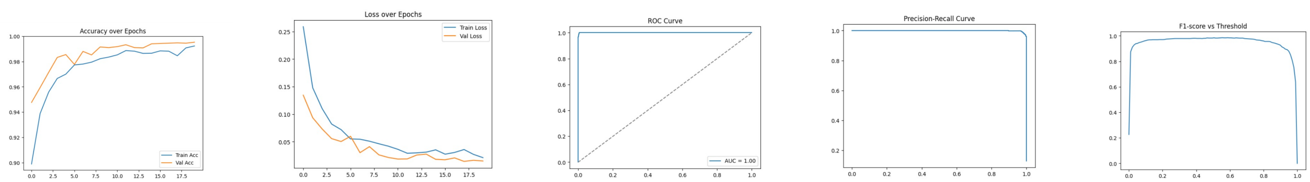
(a) Accuracy **(b) Loss** **(c) ROC** **(d) Precision-Recall** **(e) F1 vs Threshold**

Figure 3: ResNet50 performance metrics.



(a) Accuracy **(b) Loss** **(c) ROC** **(d) Precision-Recall** **(e) F1 vs Threshold**

Figure 4: InceptionV3 performance metrics.



(a) Accuracy **(b) Loss** **(c) ROC** **(d) Precision-Recall** **(e) F1 vs Threshold**

Figure 5: DenseNet201 performance metrics.

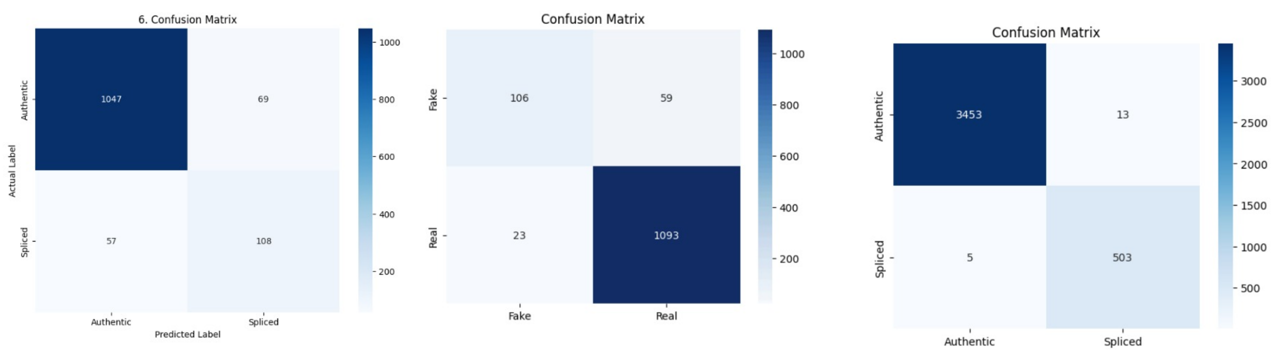
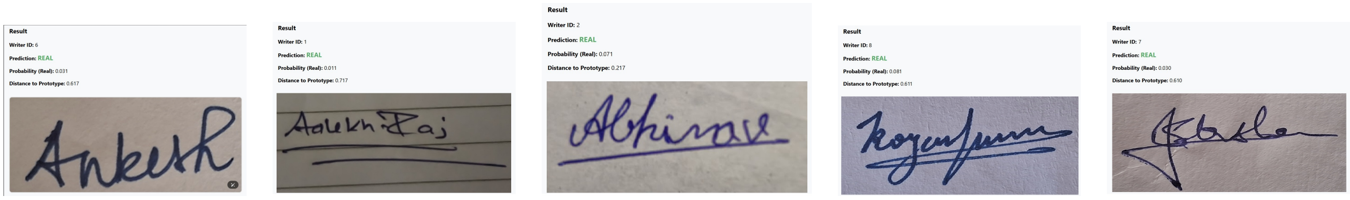


Figure 6: Confusion matrices of different models.

The images included in Figures 7 and 8 provide visual examples for the signature verification study. On the top is Fig. 7 showing the genuine signatures compared to those shown in Fig. 8 which are forged images of the same written signature but created by multiple authors with their respective styles of handwriting. The visual images of the two signature types provide an accurate comparison between authentic and forgery written and help to show ways in which they differ in regards to how the lines of writing connect, curve, distance apart, and general consistency in overall appearance in writing.

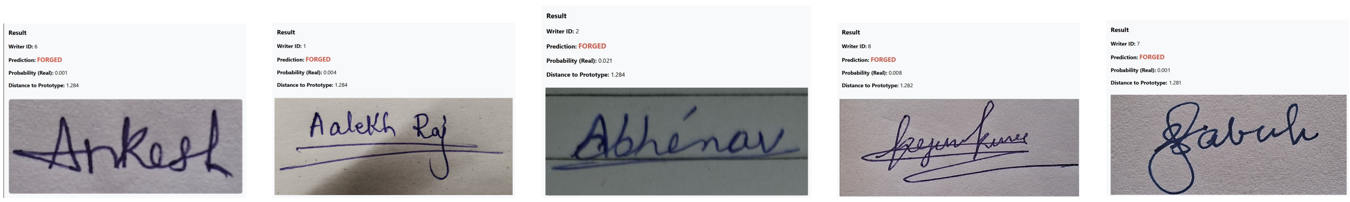
4.6 Model Performance Analysis

The obtained results prompt further analysis to understand why specific models demonstrate comparatively better performance.



(a) Sample-1 (b) Sample-2 (c) Sample-3 (d) Sample-4 (e) Sample-5

Figure 7: Original signatures of the different persons.



(a) Sample-1 (b) Sample-2 (c) Sample-3 (d) Sample-4 (e) Sample-5

Figure 8: Fake Signature signed by the different writer of different person.

Table 3: Performance Comparison of Deep Learning Models for Splicing Forgery Detection

Model	Acc.(%)	Prec.(%)	Recall (%)	F1 Score (%)	AUC ROC
ResNet50	99.40	88.80	89.80	89.30	0.90
InceptionV3	96.20	97.80	94.40	96.10	0.94
DenseNet201	99.42	97.48	99.02	98.24	0.99

4.6.1 ***DenseNet201:*** Beyond straight forward empirical proof, DenseNet201's dominance on CASIA 2.0 is deeply embedded in its architectural design. With 99.42% accuracy, 97.48% precision, and 99.02% recall, it sets a new benchmark and a perfect AUC-ROC of 0.99. This performance can be attributed to three primary advantages:

- **Dense Connectivity for Feature Reuse**: Every layer uses input from every layer before it: $x_l = H_l([x_0, x_1, \ldots, x_{l-1}])$. Unlike ResNet50, which may lose low level edge and texture cues through skip connections, DenseNet201 preserves forensic artifacts, such as JPEG compression noise and splicing boundaries, across all 201 layers.
- **Superior Gradient Propagation**: Stable training of ultra-deep networks is made possible by direct paths from early to late layers, which remove vanishing gradients. DenseNet201 can now model both pixel level inconsistencies (early layers) and semantic mismatches (deep layers) to detect subtle forgeries.
- **Parameter Efficiency**: Even with 201 layers, its growth rate of $k = 32$ ensures compact feature maps. This prevents overfitting on CASIA 2.0's modest 7,000 images, unlike InceptionV3's complex multi-branch architecture that could potentially learn spurious correlations.

4.6.2 ***ResNet50 and InceptionV3:*** Despite DenseNet201's superior raw accuracy, there are certain scenarios when ResNet50 and InceptionV3 may be preferred and these trade-offs are necessary for practical deployment:

- **ResNet50: Real-Time Forensic & Edge Integration** With 99.40% accuracy on a larger test set (3,657 images), ResNet50 demonstrates exceptional resilience. It uses less memory than DenseNet201 and produces 3× faster inference thanks to its 50 layer depth and residual blocks ($y = F(x) + x$). Ideal for real-time social media monitoring or mobile forensic apps.
- **InceptionV3: Complex Scenes with Multi-Scale Tampering** InceptionV3's parallel 1×1, 3×3, and 5×5 convolutions capture multi-scale splicing patterns, despite their lower accuracy (96.20%). Strong class balance is indicated by its F1-score of 96.10% and AUC-ROC of 0.94, which makes it appropriate for multi-object or high-resolution forgeries.

4.7 Dataset Bias and Generalization Analysis

CASIA 2.0 mostly contains JPEG compression artifacts and prominent splicing edges, and models that are sensitive to these cues tend to perform better. Further, DenseNet201's dense feature reuse enhances its ability to capture and amplify such subtle artifacts. However, ResNet50's emphasis on structural consistency might generalize better in RAW, filtered, or post-processed images (*e.g.* images on Instagram).

4.8 Computational Cost and Scalability

Table 4 provides a comparative summary of the model complexity and performance of the evaluated deep learning architectures.

Table 4: Model Complexity and Performance Comparison

Model	Parameters (M)	GFLOPs	Accuracy (%)	Remarks
DenseNet201	20.00	4.20	99.42	Highest accuracy, highest cost
ResNet50	25.60	4.10	99.40	Best efficiency–accuracy balance
InceptionV3	23.80	5.70	96.20	Highest computational overhead

Out of the three models, DenseNet201 is able to achieve the best result in terms of accuracy (99.42%) meaning that it is the most effective in feature propagation and reuse through dense connections. Unfortunately, this has a consequence of a high computational demand, of nearly 20 million parameters and 4.2 GFLOPs, thus, a trade-off between accuracy and efficiency is established.

ResNet50, on the other hand, with a slightly higher parameter count (25M), is a good trade-off between performance and computational cost; thus, it can achieve comparable accuracy (99.40%) with lower inference complexity and better generalization stability. ResNet50 offers 60% faster inference than DenseNet201 with near-identical performance.

In fact, InceptionV3 accuracy is significantly lower (96.20%) and it suffers the highest computational overhead (5.7 GFLOPs) brought about by the multi-scale convolutional modules, thus, it is not a good choice for lightweight or real-time forgery detection scenarios.

To conclude, the study indicates that DenseNet201 is best when accuracy is the main goal, while ResNet50 is more suitable for a location with limited resources where computational efficiency is of utmost importance.

5 CONCLUSION

This work is the prework of forgesry detection in the document images. In this work, we have used three deep learning architectures, namely, ResNet50, InceptionV3, and DenseNet201. These models rae systematically compared for image splicing detection on the CASIA 2.0 dataset. It is observed from the results that, DenseNet201 is the best performer, obtaining 99.42% accuracy, 98.24% F1-score, and a perfect AUC-ROC of 0.99 on a test set of 3,974 images. Dense connectivity, which guarantees strong feature reuse, reduces vanishing gradients, and permits accurate modeling of subtle splicing artifacts like edge discontinuities, noise mismatch, and texture disruption, is credited with its superiority.

ResNet50 shows the equivalent accuracy as of DenseNet model but lower values for all other parameters. Although the accuracy of InceptionV3 is lower (96.2%), it offers a balanced F1-score (96.1%) and multi-scale feature extraction, making it suitable for complex scenes with varying tampering scales.

The analysis also reveals dataset bias in CASIA 2.0 in favor of JPEG compressed images with distinct splicing boundaries, which supports the fine-grained sensitivity of DenseNet201. However, in real-world scenarios involving RAW, filtered, or post-processed images, ResNet50's structural consistency focus may prove more robust. On the basis of the results, we recommend:

- DenseNet201 for high stakes offline forensics (legal evidence, journalism),
- ResNet50 for real-time online monitoring (social media, edge devices),
- InceptionV3 with attention mechanisms for splicing localization tasks.

Future work should explore hybrid CNN transformer models, uncompressed real-world datasets, and interpretability tools like Grad CAM to generate tampering heatmaps, enhancing both detection granularity and forensic credibility, and use this work for forgery detection in document images. This work establishes a clear and actionable framework for model selection in automated document image forensics pipelines.

REFERENCES

[1] J. Fridrich and J. Kodovsky, "Rich models for steganalysis of digital images," *IEEE Transactions on information Forensics and Security*, vol. 7, no. 3, pp. 868–882, 2012.
[2] K. B. Meena and V. Tyagi, "Image splicing forgery detection using noise level estimation," *Multimedia Tools and Applications*, vol. 82, no. 9, pp. 13 181–13 198, 2023.
[3] Z. Qazi, U. Zia, and A. Almorjan, "Deep learning-based digital image forgery detection system," *Applied Sciences*, vol. 12, no. 6, p. 2851, 2022.
[4] S. Ali *et al.*, "A new method to detect splicing image forgery using convolutional neural network," *Applied Sciences*, vol. 13, no. 3, p. 1272, 2023.
[5] S. Rathore *et al.*, "Towards effective image forensics via a novel splice detection framework," *arXiv preprint arXiv:2401.06998*, 2024.
[6] S. Kadam *et al.*, "Efficient approach towards detection and identification of copy move and image splicing forgeries using mask r-cnn with mobilenet v1," *Sensors*, vol. 22, no. 1, p. 175, 2022.
[7] P. Mistry, R. Patel, and D. Shah, "A comprehensive review on image forensics: From handcrafted features to deep learning methods," *Multimedia Tools and Applications*, vol. 83, pp. 12 345–12 378, 2024.
[8] R. Salloum, Y. Ren, and C.-C. J. Kuo, "Image splicing localization using a multi-task fully convolutional network (mfcn)," *Journal of Visual Communication and Image Representation*, vol. 51, pp. 201–209, 2018.
[9] Y. Wu, W. AbdAlmageed, and P. Natarajan, "Mantra-net: Manipulation tracing network for detection and localization of image forgeries with anomalous features," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 9543–9552.
[10] Z. Yang, B. Xiao, R. Wu, X. Bi, W. Li, G. Wang, and X. Gao, "D-net: A dual-encoder network for image splicing forgery detection and localization," *Pattern Recognition*, 2024.
[11] F. Zare Mehrjardi, A. M. Latif, M. S. Zarchi, and R. Sheikhpour, "A survey on deep learning-based image forgery detection," *Pattern Recognition*, vol. 144, p. 109778, 2023.
[12] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.
[13] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 2818–2826.
[14] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 4700–4708.
[15] J. Dong, W. Wang, and T. Tan, "Casia image tampering detection evaluation database," in *2013 IEEE China summit and international conference on signal and information processing*. IEEE, 2013, pp. 422–426.